

Q1. Draw the block diagram of a standard negative-feedback control system and identify its major components.

Q2. Give five examples of:

- (a) open-loop control system,
- (b) closed-loop control system.

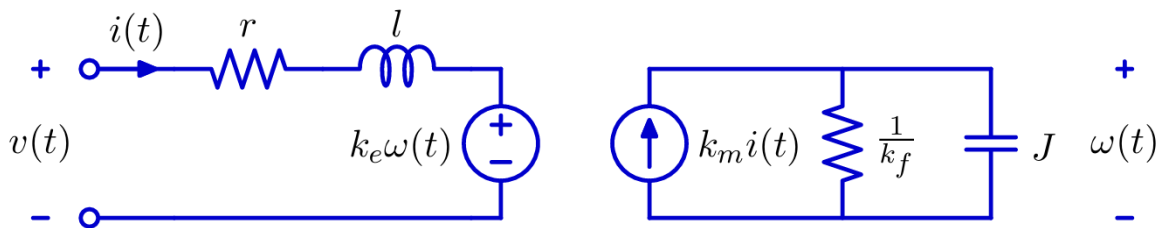
and draw their block diagrams.

Q3. Consider a system governed by the differential equation

$$\frac{d^2y(t)}{dt^2} + 5\frac{dy(t)}{dt} + 6y(t) = 2u(t).$$

Obtain its state-space model.

Q4. Consider the DC motor model. The voltage $v(t)$ represents the electrical input to the motor. It excites a current $i(t)$, which generates a torque $k_m i(t)$ that tends to rotate the motor shaft. The torque is resisted by the moment of inertia J and by friction (k_f). As the motor spins, it generates a back emf ($k_e \omega(t)$) that tends to reduce the electrical current $i(t)$ drawn by the motor.



The mathematical description of the model is as follows:

Electrical port:

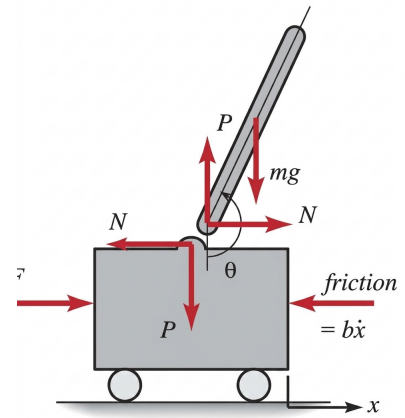
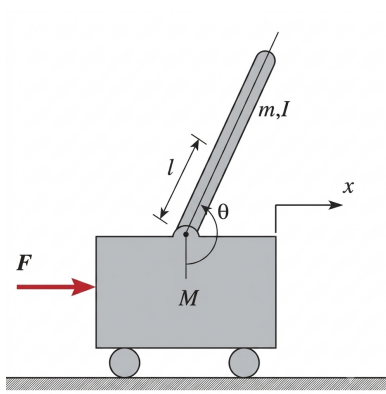
$$v(t) = ri(t) + l\frac{di(t)}{dt} + k_e\omega(t)$$

Mechanical port:

$$k_m i(t) = k_f \omega(t) + J\frac{d\omega(t)}{dt}$$

Determine a state-space description of this system.

Q5. Consider the inverted pendulum mounted on a cart. The cart has mass M , and the pendulum has mass m , moment of inertia I , and length l measured from the pivot to the center of mass. A horizontal force F is applied to the cart. The horizontal displacement of the cart is denoted by $x(t)$, and the angular displacement of the pendulum from the vertical direction is denoted by $\theta(t)$. The viscous friction coefficient of the cart is b .



The objective is to obtain a linearized state-space model of the cart-pendulum system about the upright equilibrium position.

The nonlinear equations of motion of the system are given as follows:

Horizontal motion:

$$(M + m)\ddot{x}(t) + b\dot{x}(t) + ml\ddot{\theta}(t) \cos \theta(t) - ml\dot{\theta}^2(t) \sin \theta(t) = F(t)$$

Rotational motion of the pendulum:

$$(I + ml^2)\ddot{\theta}(t) + mgl \sin \theta(t) = -ml\ddot{x}(t) \cos \theta(t)$$

Assume that the system operates in a small neighbourhood of the upright equilibrium position

$$\theta = \pi.$$

For small deviations $\phi(t)$ about the upright position, let

$$\theta(t) = \pi + \phi(t),$$

where $\phi(t)$ is small. Hence,

$$\cos(\pi + \phi) \approx -1, \quad \sin(\pi + \phi) \approx -\phi, \quad \dot{\theta}^2 \approx 0.$$

Using these approximations, the linearized equations of motion are

$$(I + ml^2)\ddot{\phi}(t) + mgl\phi(t) = ml\ddot{x}(t),$$

$$(M + m)\ddot{x}(t) + b\dot{x}(t) - ml\ddot{\phi}(t) = F(t).$$

Determine the state-space description of this system by choosing the state variables as

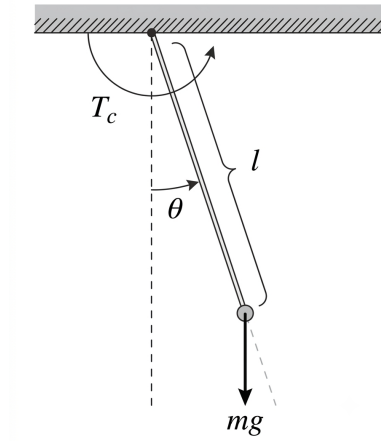
$$x_1 = x(t), \quad x_2 = \dot{x}(t), \quad x_3 = \phi(t), \quad x_4 = \dot{\phi}(t),$$

and input

$$u(t) = F(t).$$

Q6. Consider a simple pendulum system. The pendulum has mass m , length l , and moment of inertia I . A control torque $T_c(t)$ is applied at the pivot. The angular displacement of

the pendulum from the vertical direction is denoted by $\theta(t)$. The damping coefficient is b .



Using Newton's law for rotational motion,

$$F = I\alpha,$$

or equivalently,

$$I\alpha = -lmg \sin \theta - b\omega + T_c,$$

where

$$\omega = \dot{\theta}, \quad \alpha = \dot{\omega} = \ddot{\theta},$$

and

$$I = ml^2.$$

Therefore, the equation of motion of the pendulum is

$$ml^2\ddot{\theta}(t) = -lmg \sin \theta(t) - b\dot{\theta}(t) + T_c(t).$$

Equivalently,

$$\ddot{\theta}(t) = -\frac{b}{ml^2}\dot{\theta}(t) - \frac{g}{l} \sin \theta(t) + \frac{T_c(t)}{ml^2}.$$

Assume that there is no control torque, i.e.,

$$T_c(t) = 0.$$

Determine the state-space description of this system.